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RESEARCH	Spatiotemporal Machine Learning, Deep Learning, AI for Science. <i>Google Scholar:</i> https://scholar.google.com/citations?user=4HTITaMAAAAJ&hl=en	
CURRENT APPOINTMENT	University of California, San Diego , La Jolla, California <i>Professor, Department of Computer Science and Engineering</i>	July 2026 - Present
PREVIOUS APPOINTMENT	University of California, San Diego , La Jolla, California <i>Associate Professor, Department of Computer Science and Engineering</i>	July 2024 - June 2026
	<i>Assistant Professor, Department of Computer Science and Engineering</i>	July 2020 - June 2024
	Amazon Fulfillment Technologies , Remote/Bellevue, Washington <i>Amazon Scholar, AFT AI Team</i>	June 2025 - June 2026
	Google Cloud , Remote/Sunnyvale, California <i>Visiting Researcher, Cloud AI Research</i>	Oct 2021 - March 2023
	Northeastern University , Boston, Massachusetts <i>Assistant Professor, Khoury College of Computer Sciences</i> <i>Network Science Institute, College of Engineering, Physics (By Courtesy)</i>	Aug 2018 - June 2020
	California Institute of Technology , Pasadena, California <i>Postdoctoral Scholar, Computing + Mathematical Sciences</i> Advisors: Anima Anandkumar, Yisong Yue	Aug 2017 - Aug 2018
	Stanford University , Palo Alto, California <i>Visiting Researcher, Computer Science Department</i> Host: Christopher Ré	Aug 2016 - Oct 2016
	IBM Research , Yorktown Heights, New York <i>Research Intern</i>	June 2015 - Aug 2015
	Yahoo! Labs , Sunnyvale, California <i>Research Intern</i>	June 2014 - Aug 2014
	Intel Lab , Santa Clara, California <i>Research Intern</i>	May 2013 - Aug 2013
EDUCATION	University of Southern California , Los Angeles, California <i>Ph.D., Computer Science, Computer Science Department</i> <i>Finalist in William F. Ballhaus, Jr. Prize for Excellence in Graduate Engineering Research</i> Thesis Committee: Yan Liu (chair), Cyrus Shahabi (co-chair), Mahdi Soltanolkotabi	Aug 2012 - Aug 2017
	Zhejiang University , Hangzhou, Zhejiang, PRC <i>B.S in Computer Science, Chu Kochen Honors College</i> <i>Outstanding graduate of Zhejiang Province</i> Advisor: Zhihua Zhang	Aug 2008 - June 2012

SELECTED AWARDS AND HONORS	Amazon Research Award , Amazon	Feb 2026
	Samsung AI Researcher of the Year , Samsung Electronics	Sep 2025
	Presidential Early Career Award for Scientists and Engineers (PECASE)	Jan 2025
	Google Academic Research Award , Google Research	Dec 2025
	Innovators Under 35 , MIT Technology Review	Sep 2024
	Best Paper Award , ICML ML for Earth System Modeling Workshop	July 2024
	Young Faculty Award , DARPA	June 2024
	Outstanding Datasets and Benchmarks Paper Award , NeurIPS	Dec 2023
	Early Career Awards for Scientists and Engineers , Army Research Office	May 2023
	Faculty Research Award , NEC Laboratories America	March 2023
	NSF CAREER Award , National Science Foundation	June 2022
	Hellman Fellow , Hellman Fellows Fund	May 2022
	Outstanding Faculty Researcher Award , JPMorgan	July 2021
	Facebook Data Science Research Awards , Facebook Research	June 2021
	AWS Machine Learning Research Awards , Amazon Science	Feb 2021
	Best Paper Award , NeurIPS ML for Public Health Workshop	Dec 2020
	Adobe Data Science Research Awards , Adobe Research	March 2020
	Google Faculty Research Award , Google Research	Feb 2020
	Best Dissertation Award , University of Southern California	Aug 2018
	Best Paper Award , NIPS Time Series Workshop	Dec 2017
	Rising Stars in EECS , 40 Awardees in North America, MIT	Nov 2015
	Annenberg Graduate Fellowship , University of Southern California	Aug 2012
	Selected in ACM Heidelberg Laureate Forum , University of Heidelberg	Sep 2013
	Microsoft Young Scholar Fellowship , Microsoft Research Asia	Aug 2011
	International Forum (iF) Design Hanover Global Concept Award , iF	June 2010
	First prize in Undergraduate Research and Innovation , Zhejiang University	Sep 2010

CONFERENCE
PUBLICATIONS

[C1] Pratham Yashwante, [Rose Yu](#). “Time Series, Vision, and Language: Exploring the Limits of

Alignment in Contrastive Representation Spaces.” To Appear in ICML 2026

- [C2] Salva Rühling Cachay, Duncan Watson-Parris, [Rose Yu](#). “U-Cast: A Surprisingly Simple Frontier Probabilistic AI Weather Forecaster.” To Appear in ICML 2026
- [C3] Yadi Cao, Sicheng Lai, Jiahe Huang, Yang Zhang, Zach Lawrence, Rohan Bhakta, Izzy F. Thomas, Mingyun Cao, Chung-Hao Tsai, Zihao Zhou, Yidong Zhao, Hao Liu, Alessandro Marinoni, Alexey Arefiev, [Rose Yu](#). “SimulCost: A Cost-Aware Benchmark and Toolkit for Automating Physics Simulations with LLMs.” To Appear in ICML 2026.
- [C4] Luca Zhou, Bo Zhao, [Rose Yu](#), Emanuele Rodolà. “Demystifying Mergeability: Interpretable Properties to Predict Model Merging Success.” To Appear in ICML 2026
- [C5] Esha Singh, Dongxia Wu, Chien-Yi Yang, Tajana Rosing, [Rose Yu](#), Yian Ma. “Divide and Learn: Multi-Objective Combinatorial Optimization at Scale.” To Appear in ICML 2026
- [C6] Ian Li, Zilei Shao, Benjie Wang, [Rose Yu](#), Guy Van den Broeck, Anji Liu. “Breaking the Factorization Barrier in Diffusion Language Models.” To Appear in ICML 2026
- [C7] Maya Okawa, Bo Zhao, Eric Bigelow, [Rose Yu](#), Tomer Ullman, Ekdeep Singh Lubana, Hidenori Tanaka. “Emergence of Hierarchical Emotion Organization in Large Language Models.” To Appear in ICML 2026
- [C8] Varambally, Sumanth, Thomas Voice, Yanchao Sun, Zhifeng Chen, [Rose Yu](#), and Ke Ye. “Hilbert: Recursively building formal proofs with informal reasoning.” ICLR 2026.
- [C9] Sumanth Varambally, Marshall Fisher, Jas Thakker, Yiwei Chen, Zhirui Xia, Yasaman Jafari, Ruijia Niu, Manas Jain, Veeramakali Vignesh Manivannan, Zachary Novack, Luyu Han, Srikar Eranky, Salva Rühling Cachay, Taylor Berg-Kirkpatrick, Duncan Watson-Parris, Yi-An Ma, [Rose Yu](#). “Zephyrus: An Agentic Framework for Weather Science” ICLR 2026.
- [C10] Nicholas C King, Jared Maeyama, Shubh Maheshwari, Andrew McCulloch, [Rose Yu](#). “PG-BIG: Personalized Guidance for Biomechanically Informed Generative Models in Exercise.” L4DC 2026
- [C11] Salva Rühling Cachay, Miika Aittala, Karsten Kreis, Noah Brenowitz, Arash Vahdat, Morteza Mardani, [Rose Yu](#). “Elucidated Rolling Diffusion Models for Probabilistic Forecasting of Complex Dynamics.” NeurIPS 2025
- [C12] Sophia Sun, [Rose Yu](#) “Conformal Prediction for Time-series Forecasting with Change Points.” NeurIPS 2025
- [C13] Bohan Lyu, Yadi Cao, Duncan Watson-Parris, Leon Bergen, Taylor Berg-Kirkpatrick, [Rose Yu](#). “Adapting While Learning: Grounding LLMs for Scientific Problems with Tool Usage Adaptation.” ICML 2025
- [C14] Peter Eckmann, Dongxia Wu, Germano Heinzemann, Michael K Gilson, [Rose Yu](#). “MF-LAL: Drug Compound Generation Using Multi-Fidelity Latent Space Active Learning.” ICML 2025
- [C15] Kun Wang, Sumanth Varambally, Duncan Watson-Parris, Yian Ma, [Rose Yu](#). “Discovering Latent Structural Causal Models from Spatiotemporal Data”. ICML 2025
- [C16] Bo Zhao, Nima Dehmamy, Robin Walters, [Rose Yu](#). “Understanding Mode Connectivity via Parameter Space Symmetry” ICML 2025
- [C17] Manu Bhat, Jonghyun Park, Jianke Yang, Nima Dehmamy, Robin Walters, [Rose Yu](#). “AtlasD: Automatic Local Symmetry Discovery.” ICML 2025
- [C18] Zihao Zhou, [Rose Yu](#). “Can LLMs Understand Time Series Anomalies?” ICLR 2025
- [C19] Veeramakali Vignesh Manivannan, Yasaman Jafari, Srikar Eranky, Spencer Ho, [Rose Yu](#), Duncan Watson-Parris, Yian Ma, Leon Bergen, Taylor Berg-Kirkpatrick. “ClimaQA: An Automated Evaluation Framework for Climate Question Answering Models.” ICLR 2025
- [C20] Shubh Maheshwari, Anwesh Mohanty, Yadi Cao, Swithin Razu, Andrew McCulloch, [Rose Yu](#). “BIGE : Biomechanics-informed GenAI for Exercise Science.” L4DC 2025

- [C21] Salva Rühling Cachay, Brian Henn, Oliver Watt-Meyer, Christopher S. Bretherton, [Rose Yu](#). “Probabilistic Emulation of a Global Climate Model with Spherical DYffusion.” NeurIPS 2024 **Spotlight**
- [C22] Jianke Yang, Wang Rao, Nima Dehmamy, Robin Walters, [Rose Yu](#). “Symmetry-Informed Governing Equation Discovery.” NeurIPS 2024
- [C23] Csaba Both, Nima Dehmamy, Jianzhi Long, [Rose Yu](#). “Faster Optimization on Sparse Graphs via Neural Reparametrization.” In *Learning on Graphs Conference*(LoG), 2024
- [C24] Sumanth Varambally, Yian Ma, [Rose Yu](#). “Discovering Mixtures of Structural Causal Models from Time Series Data.” ICML 2024
- [C25] Jianke Yang, Nima Dehmamy, Robin Walters, [Rose Yu](#). “Latent Space Symmetry Discovery.” ICML 2024
- [C26] Ruijia Niu, Dongxia Wu, Kai Kim, Duncan Watson-Parris, Yian Ma, [Rose Yu](#). “Multi-Fidelity Residual Neural Processes for Scalable Surrogate Modeling.” ICML 2024
- [C27] Tao Wang, Bo Zhao, Sicun Gao, [Rose Yu](#). “Understanding the Difficulty of Solving Cauchy Problems with PINNs.” L4DC 2024
- [C28] Sophia Sun, Wenyan Chen, Zihao Zhou, Sonia Fereidooni, Elise Jortberg, [Rose Yu](#). “Data-Driven Simulator for Mechanical Circulatory Support with Domain Adversarial Neural Process.” L4DC 2024
- [C29] Dongxia Wu, Tsuyoshi Idé, Aurélie Lozano, Georgios Kollias, Jiří Navrátil, Naoki Abe, Yian Ma, [Rose Yu](#). “Learning Granger Causality from Instance-wise Self-attentive Hawkes Processes.” AISTATS 2024
- [C30] Cai Zhou, [Rose Yu](#), Yusu Wang. “On the Theoretical Expressive Power and Design Space of High Order Graph Transformers.” AISTATS 2024
- [C31] Sophia Sun, [Rose Yu](#). “Copula Conformal Prediction for Multi-step Time Series Forecasting.” ICLR 2024
- [C32] Bo Zhao, Robert M. Gower, Robin Walters, [Rose Yu](#). “Improving Convergence and Generalization Using Parameter Symmetries.” ICLR 2024 **Oral**
- [C33] Zihao Zhou, [Rose Yu](#). “Automatic Integration for Spatiotemporal Neural Point Processes.” NeurIPS 2023
- [C34] Salva Rühling Cachay, Bo Zhao, Hailey James, [Rose Yu](#). “DYffusion: A Dynamics-informed Diffusion Model for Spatiotemporal Forecasting.” NeurIPS 2023
- [C35] Sungduk Yu, et al. “ClimSim: An open large-scale dataset for training high-resolution physics emulators in hybrid multi-scale climate models.” NeurIPS Datasets and Benchmarks, 2023 **Outstanding Paper Award**
- [C36] Dongxia Wu, Ruijia Niu, Matteo Chinazzi, Alessandro Vespignani, Yi-An Ma, [Rose Yu](#). “Deep Bayesian Active Learning for Accelerating Stochastic Simulation.” KDD 2023
- [C37] Jianke Yang, Robin Walters, Nima Dehmamy, [Rose Yu](#). “Generative Adversarial Symmetry Discovery.” ICML 2023
- [C38] Dongxia Wu, Ruijia Niu, Matteo Chinazzi, Yi-An Ma, [Rose Yu](#). “Disentangled Multi-Fidelity Deep Bayesian Active Learning.” ICML 2023
- [C39] Chen Cai, Truong Son Hy, [Rose Yu](#), Yusu Wang. “On the Connection Between MPNN and Graph Transformer.” ICML 2023
- [C40] Zihao Zhou, [Rose Yu](#). “Automatic Integration for Fast and Interpretable Neural Point Processes.” L4DC 2023
- [C41] Sophia Sun, Robin Walters, Jinxi Li, [Rose Yu](#). “Probabilistic Symmetry for Multi-Agent Dynamics.” L4DC 2023

- [C42] Rui Wang, Yihe Dong, Serkan Ö. Arik, [Rose Yu](#). "Koopman Neural Forecaster for Time Series with Temporal Distribution Shifts." ICLR 2023
- [C43] Bo Zhao, Iordan Ganev, Robin Walters, [Rose Yu](#), Nima Dehmamy. "Symmetries, flat minima, and the conserved quantities of gradient flow." ICLR 2023
- [C44] Bo Zhao, Nima Dehmamy, Robin Walters, [Rose Yu](#) "Symmetry Teleportation for Accelerated Optimization." NeurIPS 2022
- [C45] Rui Wang, Robin Walters, [Rose Yu](#) "Meta-Learning Dynamics Forecasting Using Task Inference." NeurIPS 2022
- [C46] Dongxia Wu, Matteo Chinazzi, Alessandro Vespignani, Yi-An Ma, [Rose Yu](#) "Multi-fidelity Hierarchical Neural Processes." KDD 2022
- [C47] Rui Wang, Robin Walters, [Rose Yu](#) "Approximately Equivariant Networks for Imperfectly Symmetric Dynamics." ICML 2022
- [C48] Peter Eckmann, Kunyang Sun, Bo Zhao, Mudong Feng, Michael Gilson, [Rose Yu](#) "LIMO: Latent Inceptionism for Targeted Molecule Generation." ICML 2022
- [C49] Zihao Zhou, Xingyi Yang, Ryan Rossi, Handong Zhao, [Rose Yu](#). "Neural Point Process for Learning Spatiotemporal Event Dynamics." L4DC 2022
- [C50] Alan Li, Zihao Zhou, Elise Jortbeg, [Rose Yu](#). "Generalization of Deep Sequence Models for Forecasting Aortic Pressure Cross-Cohort." In *Computing in Cardiology (CinC)*, 2022.
- [C51] Alejandro Rodriguez Pascual, Ishan Mehta, Muhammad Khan, Frank Rodriz, [Rose Yu](#). "Understanding why shooters shoot - An AI-powered engine for basketball performance profiling." MIT Sloan Sports Analytics Conference (SSAC), 2022
- [C52] Nima Dehmamy, Robin Walters, Yanchen Liu, Dashun Wang, [Rose Yu](#). "Automatic Symmetry Discovery with Lie Algebra Convolutional Network." NeurIPS 2021
- [C53] Dongxia Wu, Liyao Gao, Xinyue Xiong, Matteo Chinazzi, Alessandro Vespignani, Yi-An Ma, [Rose Yu](#). "Quantifying Uncertainty in Deep Spatiotemporal Forecasting." KDD 2021
- [C54] Rui Wang, Danielle Maddix, Christos Faloutsos, Yuyang Wang, [Rose Yu](#). "Bridging Physics-based and Data-driven modeling for Learning Dynamical Systems." L4DC 2021
- [C55] Steven Wong, Lejun Jiang, Robin Walters, Tamás G. Molnár, Gábor Orosz, [Rose Yu](#). "Physics-Guided Deep Learning for Traffic Forecasting using Vehicle-to-Vehicle Communication." L4DC 2021
- [C56] Robin Walters, Jinxi (Leo) Li, [Rose Yu](#). "Trajectory Prediction using Equivariant Continuous Convolution." ICLR 2021
- [C57] Rui Wang, Robin Walters, [Rose Yu](#). "Incorporating Symmetry into Deep Dynamics Models for Improved Generalization." ICLR 2021
- [C58] Fan Xie, Alex Chowdhury, Clara De Paolis, Linfeng Zhao, Lawson Wong, [Rose Yu](#). "Deep Imitation Learning for Bimanual Robotic Manipulation." NeurIPS 2020
- [C59] Armand Comas Massague, Chi Zhang, Zlatan Feric, Octavia Camps, [Rose Yu](#). "Learning Disentangled Representations of Video with Missing Data." NeurIPS 2020
- [C60] Jung Yeon Park, Kenneth Theo Carr, Stephan Zheng, Yisong Yue, [Rose Yu](#) "Multiresolution Tensor Learning for Efficient and Interpretable Spatial Analysis." ICML 2020
- [C61] Chintan Shah, Nima Dehmamy, Nicola Perra, Matteo Chinazzi, Albert-László Barabási, Alessandro Vespignani, [Rose Yu](#). "Finding Patient Zero: Learning Contagion Source with Graph Neural Networks." In *International Conference on Network Science (NetSci)*, 2020.
- [C62] Csaba Both, Nima Dehmamy, Albert-László Barabási, [Rose Yu](#). "Network Layout using Graph Neural Networks." NetSci 2020.
- [C63] Rui Wang, Adrian Albert, Karthik, Kashinath, Mustafa, Mustafa, [Rose Yu](#). "Towards Physics-informed Deep Learning for Spatiotemporal Modeling of Turbulent Flows." KDD 2020

- [C64] Eliza Huang, Rui Wang, Uma Chandrasekaran, Rose Yu. "Aortic Pressure Forecasting with Deep Sequence Learning", In Proceeding of *Computing in Cardiology* (CinC), 2020
- [C65] Nima Dehmamy, Albert-László Barabási, Rose Yu. "Understanding the Representation Power of Graph Neural Networks in Learning Graph Topology." NeurIPS 2019
- [C66] Yukai Liu, Rose Yu, Stephan Zheng, Eric Zhan, Yisong Yue. "NAOMI: Non-Autoregressive Multiresolution Sequence Imputation." NeurIPS 2019
- [C67] Guanya Shi, Xichen Shi, Michael O'Connell, Rose Yu, Kamyar Azizzadenesheli, Anima Anandkumar, Yisong Yue, Soon-Jo Chung. "Neural Lander: Stable Drone Landing Control using Learned Dynamics", ICRA 2019
- [C68] Yaguang Li, Rose Yu, Cyrus Shahabi, Yan Liu. "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting", ICLR 2018
- [C69] Rose Yu, Guangyu Li, Yan Liu. "Tensor regression meets Gaussian Processes." AISTATS 2018
- [C70] Rose Yu, Yaguang Li, Ugur Demiryurek, Cyrus Shahabi, Yan Liu. "Deep Learning: A Generic Approach for Extreme Condition Traffic Forecasting." In Proceedings of *the Seventeenth SIAM International Conference on Data Mining* (SDM), 2017
- [C71] Rose Yu, Yan Liu. "Learning from Multiway Data: Simple and Efficient Tensor Regression." ICML 2016
- [C72] Dingxiong Deng, Cyrus Shahabi, Ugur Demiryurek, Linhong Zhu, Rose Yu, Yan Liu, "Latent Space Model for Road Networks to Predict Time-Varying Traffic", KDD 2016
- [C73] Rose Yu, Andrew Gelfand, Suju Rajan, Cyrus Shahabi, Yan Liu. "Geographic Segmentation via Latent Poisson Factor Model." In *ACM International Conference on Web Search and Data Mining* (WSDM), 2016
- [C74] Rose Yu, Dehua Cheng, Yan Liu. "Accelerated Online Low Rank Tensor Learning for Multivariate Spatiotemporal Streams." ICML 2015
- [C75] Rose Yu, Mohammad Taha Bahadori, Yan Liu. "Fast Multivariate Spatio-temporal Analysis via Low Rank Tensor Learning." NIPS 2014 **Spotlight**
- [C76] Rose Yu, Xinran He, Yan Liu. "GLAD: Group Anomaly Detection in Social Media Analysis." In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (KDD), 2014
- [C77] Cuixia Gao, Naiyan Wang, Qi Yu, Zhihua Zhang. "A Feasible Nonconvex Relaxation Approach to Feature Selection." In *Proceeding of 24th AAAI Conference on Artificial Intelligence* (AAAI), 2011

JOURNAL
PUBLICATIONS

- [J1] Thakkar, Nitya, Mert Yuksekgonul, Jake Silberg, Animesh Garg, Nanyun Peng, Fei Sha, Rose Yu, Carl Vondrick, and James Zou. "A large-scale randomized study of large language model feedback in peer review." *Nature Machine Intelligence* (2026): 1-11.
- [J2] Cao, Yadi, Yuxuan Liu, Liu Yang, Rose Yu, Hayden Schaeffer, and Stanley Osher. "Vicon: Vision in-context operator networks for multi-physics fluid dynamics prediction." *Transactions on Machine Learning Research* (TMLR) 2026.
- [J3] Eckmann, Peter, Jake Anderson, Rose Yu, and Michael K. Gilson. "Ligand-Based Compound Activity Prediction via Few-Shot Learning." *Journal of Chemical Information and Modeling* 64, no. 14 (2024): 5492-5499.
- [J4] Lee, Audrey, et al. "Preliminary Validity and Acceptability of Motion Tape for Measuring Low Back Movement: Mixed Methods Study." *JMIR Rehabilitation and Assistive Technologies* 11.1 (2024): e57953.
- [J5] Eyring, Veronika, et al. "Pushing the frontiers in climate modelling and analysis with machine learning." *Nature Climate Change* (2024): 1-13.

- [J6] Rose Yu, Rui Wang. “Learning Dynamical Systems from Data: An Introduction to Physics-Guided Deep Learning”, In Proceedings of the National Academy of Sciences 121.27 (2024): e2311808121.
- [J7] Abhimanyu Das, Weihao Kong, Andrew Leach, Shaan Mathur, Rajat Sen, Rose Yu. “Long-term Forecasting with TiDE: Time-series Dense Encoder”, *Transactions on Machine Learning Research* (TMLR), 2023
- [J8] Krenn, Mario, et al. “Forecasting the future of artificial intelligence with machine learning-based link prediction in an exponentially growing knowledge network.” *Nature Machine Intelligence* 5.11 (2023): 1326-1335.
- [J9] Csaba Both, Nima Dehmamy, Rose Yu and Albert-László Barabási. “Accelerating network layouts using graph neural networks.” *Nature Communications* 14, no. 1 (2023): 1560.
Chatterjee, Ayan, et al. “Improving the generalizability of protein-ligand binding predictions with AI-Bind.” *Nature communications* 14.1 (2023): 1989.
- [J10] Krenn, Mario, Qianxiang Ai, Senja Barthel, Nessa Carson, Angelo Frei, Nathan C. Frey, Pascal Friederich et al. “SELFIES and the future of molecular string representations.” *Patterns* 3, no. 10 (2022).
- [J11] Utkrisht Rajkumar, Sara Javadzadeh, Mihir Bafna, Dongxia Wu, Rose Yu, Jingbo Shang, Vineet Bafna. “DeepViFi: Detecting Oncoviral Infections in Cancer Genomes using Transformers”. In *Proceedings of the 13th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, pp. 1-8. 2022.
- [J12] Cramer, E.Y., Lopez, V.K., Niemi, J., George, G.E., Cegan, J.C., Dettwiller, I.D., England, W.P., Farthing, M.W., Hunter, R.H., Lafferty, B. and Linkov, I., 2021. “Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the US”. In *Proceedings of the National Academy of Sciences* (PNAS), 2022
- [J13] K. Kashinath, M. Mustafa, A. Albert, J-L. Wu, C. Jiang, S. Esmailzadeh, K. Azizzadenesheli, R. Wang, A. Singh, A. Manepalli, D. Chirila, R. Yu, R. Walters, B. White, H. Xiao, H. A. Tchelepi, P. Marcus, A. Anandkumar, Prabhat. “Physics-informed machine learning: Case studies for weather and climate modeling” In *Journal of Philosophical Transactions of the Royal Society A*, 2020
- [J14] Rose Yu, Stephan Zheng, Anima Anandkumar, Yisong Yue. “Long Term Forecasting with Higher Order Tensor RNN”. In *Journal of Machine Learning Research* (JMLR), 2018
- [J15] Rose Yu, Yan Liu. “Spatio-Temporal Analysis of Social Media Data” In *Encyclopedia of GIS*, 2016
- [J16] Rose Yu, Huida Qiu, Zhen Wen, Ching-Yung Liu, Yan Liu. “A Survey on Social Media Analysis Anomaly Detection” In *ACM KDD Exploration*, 2016
- [J17] Rose Yu, Xinran He, Yan Liu. “GLAD: Group Anomaly Detection in Social Media Analysis - Extended Abstract.” In *ACM Transactions on Knowledge Discovery in Data* (TKDD), 2015

AWARDED GRANTS PI or Co-PI of over 62M total funding. Numbers in parenthesis are our portion.

- [G1] PI, Amazon Research Award, *AI Co-Scientist: Designing Autonomous Agents for ML Discovery*, \$150,000, Oct 2025 - Oct 2026
- [G2] Co-PI, ARPA-H, *Large-scale, Intelligent, Vascularized, Engineered, Regenerative Liver* \$25,828,992 (\$877,731) May 2025 - Dec 2029
- [G3] Co-PI, Google Academic Research Awards, *ClimateBench2.0: Probabilistic climate model scoring*, \$270,000, Oct 2024 - Sep 2025
- [G4] PI, DARPA Young Faculty Award, *Grounding LLMs with Physical Laws*, \$250,000, Sep 2024 - Aug 2026

- [G5] PI, DARPA AIE FoundSci, *GENIE: Generative Foundation Model for Automated Climate Science*, \$999,306, Jan 2024 - June 2025
- [G6] co-PI, CDC CFA, *Epistorm: Center for Advanced Epidemic Analytics and Predictive Modeling Technology*, \$17.5M (\$499,998), Oct 2023 - Sep 2028
- [G7] co-PI, DOE FES, *DE-SC0024499: A Fusion Machine Learning Data Science Platform to Support the Design and Safe Operation of a Fusion Pilot Plant*, \$7.4M (\$534,747), Aug 2023 - July 2026
- [G8] PI, IARPA HAYSTAC Program Sub-Contract: *ROMULUS and REMUS - Two Systems for Building Mega-City Intelligence*, \$642,745, June 2023 - June 2027
- [G9] PI, Army Research Office, ECASE: *W911NF23-1-0231: Exploiting symmetry for Learning Spatiotemporal Dynamics*, \$999,999, June 2023 - June 2028
- [G10] PI, NEC Faculty Research Gift Award. *Physics-Guided Generative Modeling for Molecules*, \$30,000, Mar 2023
- [G11] Co-PI, JSOE Early Career Faculty Development Award: *AI-assisted recording of intracellular action potential waveforms from heart muscle cells*, \$50,000, Jan - June 2023
- [G12] Co-PI, MRPI 2023 Planning/Pilot Awards: UC Collaborative for AI-enabled Materials Exploration and Optimization (UC-CAMEO), \$299,912, Jan 2023 - Jan 2025
- [G13] Co-PI, NSF IIS *2205093: SCH: MS-ADAPT: Multi-Sensor Adaptive Data Analytics for Physical Therapy*, \$1,199,930, Sep 2022 - Aug 2026
- [G14] PI, 2022 MICS Mini-Project, *Learning to Discover Causal Graphs from Multi-Agent Spatiotemporal Data*, \$17,000, Sep 2022 - Aug 2023
- [G15] PI, NSF CAREER, *2146343: New Frontiers In Large-Scale Spatiotemporal Data Analysis* \$599,999, July 2022 - June 2027
- [G16] PI, DoD DURIP, *W911NF-22-1-0179: Computational Clusters for Robotic Deep Learning in Complex Spatiotemporal Environment*, \$575,142, March 2022
- [G17] PI, NVIDIA Applied Research Accelerator Award, *Accelerating Stochastic Simulation for Fast Epidemic Scenario Creation*, 2 - A100 GPUs, March 2022
- [G18] Lead PI, DOE ASCR, *DE-SC0022255: Discovering Physically Meaningful Structures from Climate Extreme Data* \$ 3,560,504, Sep 2021 - Aug 2024
- [G19] Co-PI, NSF DMS, *2134274: SCALE MoDL: Representation Theoretic Foundations of Deep Learning*, \$300,000, Jan 2022 - Dec 2024
- [G20] Co-PI, NSF CNS, *2100237-CCRI: ABR: Cognitive Hardware and Software Ecosystem Community Infrastructure (CHASE-CI)*, \$999,971, June 2021 - May 2024
- [G21] Co-PI, NSF CNS, *2120019-CCRI: ENS: Cognitive Hardware and Software Ecosystem Community Infrastructure (CHASE-CI)*, \$1,800,000, Oct 2021 - Sep 2024
- [G22] PI, Facebook Data Science Research Award, *Sample-efficient sequential Bayesian decision making*, \$50,000, June 2021
- [G23] PI, AWS ML Research Awards, *Deep Relational Forecasting for Dynamic Graphs*, \$30,000, with \$50,000 AWS Credits, Feb, 2021
- [G24] Co-PI, DARPA SBIR, *W31P4Q-21-C-0014: Combining Simulated and Real Data for Near-Term Forecasting of Nonstationary Dynamic Processes*, \$120,000, Feb 2021 - Sep 2021.
- [G25] PI, Abiomed Research Grant, *Automated Patient Care Assistant via Machine Learning*, \$299,480, Nov 2020 - Oct 2023.
- [G26] PI, Army Research Office, *W911NF-20-1-0334: Physics-Guided Learning for Sample Efficient Spatiotemporal Decision Making*, \$370,704, Sep 2020 - Aug 2023.
- [G27] PI, Adobe Data Science Research Award, *Context-Aware Customer Journey Optimization in Spatiotemporal Environment*, \$50,000, June 2020

- [G28] PI, Google Faculty Award, *Physics Guided Deep Learning for Realistic and Efficient Traffic Simulation*, \$67,694, Feb 2020
- [G29] PI, NSF CRII, *IIS-1850349/2037745: Multiresolution Tensor Learning for Scalable and Interpretable Spatiotemporal Analysis*, \$174,998, Aug, 2019 - July, 2022
- [G30] PI, Abiomed Research Grant, *Anticipating Patient Outcome via Machine Learning*, \$50,000, July, 2019 - Sep, 2020
- [G31] PI, ONR STTR, *N68335-19-C-0310: Subcontract: Predictive Graph Convolutional Networks*, \$71,784, June, 2019 - Dec, 2019
- [G32] Co-PI, Mathworks Microgrant, *Battery State of Health Machine Learning Prognostics*, \$17,400, Sep, 2019 - Dec, 2019
- [G33] Co-PI, ONR ECP, *Subcontract: Learning and Prediction of Dynamic Processes on Evolving Networks*, \$28,840, March, 2019 - Oct, 2019
- [G34] NVIDIA GPU Grant Oct 2018, 1 Titan Xp, Amazon AWS Research Credits, Oct 2018, \$20,000 Cloud Credits, Google Cloud Research Credits, Oct, 2018, \$10,000 Cloud Credits

ADVISING

Postdocs Supervised

- Sharvaree Vadgama, 2025, active
- Yadi Cao, 2024, active
- Truong Son Hy (co-advised), 2022-2023, Assistant Professor, Indiana State University
- Jędrzej Kozerański, 2021-2022, Research Scientist, Apple
- Robin Walters, 2020-2022, Assistant Professor, Northeastern University

Ph.D. Students Advised

- Hansen Lillemark (co-advised), 2025, active
- Alex Rojas (co-advised), 2025, active
- Vivien Chen (co-advised), 2025, active
- Ian Li (co-advised), 2025, active
- Jiahe (Chloe) Huang, 2025, active
- Ruijia (Brooks) Niu (co-advised), 2024, active
- Aysin Tumay, 2024, active
- Salva Rhling Cachay, 2022, active
- Sumanth Varambally (co-advised), 2022, active
- Jianke Yang, 2022, active
- Zihao Zhou, 2021, active
- Bo Zhao, 2021, active
- Sophia Sun, 2021-2025, Applied Scientist, AWS
- Dongxia (Allen) Wu (co-advised), 2020-2025, Postdoc at Stanford
- Rui (Ray) Wang, 2019-2023, Applied Scientist, AWS

TEACHING

Deep Learning (CSE 151B & CSE 151B)	Spring 2025, 2021-2023
Generative AI/Deep Generative Models (CSE 291)	Fall 2025, 2022-2023, 2020
Deep Reinforcement Learning (CSE 291)	Fall 2021
Advanced Machine Learning (CS 7140)	Spring 2020
Introduction to Computer Science Research (CS 3950)	Fall 2019

	Special Topics in AI: Deep Learning (CS 7180)	Spring 2019
	Machine Learning (CS 6140)	Fall 2018
INVITED TALKS	Keynotes and Plenary Talks	
	ICLR Workshop: Foundation Models for Science	April 2026
	ICLR Workshop AI & PDE Workshop	April 2026
	ICLR Workshop: Time Series in the Age of Large Models	April 2026
	San Diego Engineers Week Awards Banquet	Feb, 2025
	NeurIPS AI for Science Workshop	Dec, 2025
	NeurIPS Workshop on Unifying Representations in Neural Models	Dec, 2025
	Agentic AI Summit, UC Berkeley	Aug, 2025
	Foundation Models for Sciences Conference, Flatiron Institute	April, 2025
	UC San Diego Supply Chain Forum	April, 2025
	SABPA: Frontiers in Therapeutics and Diagnostics	Mar, 2025
	Science Understanding through Data Science Conference, Caltech	Aug, 2024
	ICML Workshop on Geometry-grounded representation learning	July, 2024
	Annual Learning for Dynamics and Control Conference (L4DC)	June 2023
	MATH +X 2023 Symposium, Simons Foundation	May 2023
	NeurIPS 2022 Tackling Climate Change with ML Workshop	Dec 2022
	Physics-Informed Machine Learning Conference, Los Alamos National Lab	May 2022
	KITP Machine Learning for Climate Conference	Nov 2021
	European Research Network on System Identification	Sep, 2019
	Climate Informatics Conference 2018 (CI 2018),	Sep 2018
	Podcasts and Media	
	Into the Impossible: The Computer EXPERT That Just Solved Google's Hardest Challenge [Link]	
	Quanta Magazine: Improving Deep Learning With a Little Help From Physics [Link]	
	Carry the Two: Automatic Symmetry Discovery [Link]	
	The TWIML AI Podcast: Spatiotemporal Data Analysis [Link]	

Conference Organizing Committee

Program Chair, ICLR (2025)
Workshop Chair, NeurIPS (2024)
Industry Expo Chair, ICML (2023, 2022)
Workshop Chair, ICLR (2023)
Poster Chair, KDD (2020)
Proceedings Chair, ACM SIGSPATIAL (2020)
Coordinator, KITP: At the Crossroad of Physics and Machine Learning, 2019
Proceedings Chair, WSDM (2018)
Short Paper Chair, CIKM (2017)

Workshop Organizing Committee

ICML Time Series Workshop, (2021, 2019, 2017)
ICCV Simulation Technology for Embodied AI, 2021
ICLR Workshop on Deep Learning for Simulation (2021)
NeurIPS Workshop on Machine Learning for Engineering (2020)
NIPS Woman in Machine Learning Workshop (2016)
NIPS workshop on Learning with Tensors: Why Now and How? (2016)

Scientific Advisory and Consultancy

External Advisory Council, Allen Institute of AI, Climate Modeling Group (2024-2025)
Scientific Advisor, Salient Predictions, LLC (2023-2024)

Senior Program Committee (or Area Chair)

NeurIPS (2020-2025), ICML (2020-2025), ICLR (2021-2023, 2026), L4DC (2024), AAAI (2021), IJCAI (2020), SDM (2020)

Program Committee

ICML (2020, 2019, 2018), NeurIPS (2019, 2018), ICLR (2020, 2019), KDD (2019), AISTATS (2020, 2019), SDM (2019), AAAI (2018), IJCAI (2018), CIKM (2017), NIPS Time Series Workshop (2016), ICML Time Series Workshop (2016)

Reviewer / Panelist

National Science Foundation (NSF) Review Panel (2018-2022, 2025), Department of Energy (DOE) Review Panel (2019-2023), AAAI 2021 Spring Symposium on Combining Machine Learning with Physical Sciences, Proceedings of IEEE (2019), Journal of Machine Learning Research (JMLR), Journal of Artificial Intelligence Research (JAIR), Transactions on Knowledge Discovery from Data (TKDD), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Intelligent Transportation Systems Transaction (ITS)